

Global Stability of the Origin

Actually, for $r < 1$, we can show that *every* trajectory approaches the origin as $t \rightarrow \infty$; the origin is **globally stable**. Hence there can be no limit cycles or chaos for $r < 1$.

The proof involves the construction of a **Liapunov function**, a smooth, positive definite function that decreases along trajectories. As discussed in Section 7.2, a Liapunov function is a generalization of an energy function for a classical mechanical system—in the presence of friction or other dissipation, the energy decreases monotonically. There is no systematic way to concoct Liapunov functions, but often it is wise to try expressions involving sums of squares.

Here, consider $V(x, y, z) = \frac{1}{\sigma}x^2 + y^2 + z^2$. The surfaces of constant V are concentric ellipsoids about the origin (Figure 9.2.3).

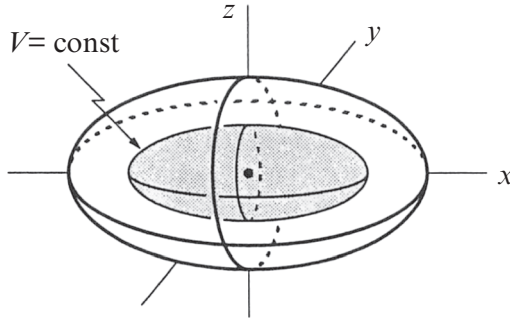


Figure 9.2.3

The idea is to show that if $r < 1$ and $(x, y, z) \neq (0, 0, 0)$, then $\dot{V} < 0$ along trajectories. This would imply that the trajectory keeps moving to lower V , and hence penetrates smaller and smaller ellipsoids as $t \rightarrow \infty$. But V is bounded below by 0, so $V(\mathbf{x}(t)) \rightarrow 0$ and hence $\mathbf{x}(t) \rightarrow \mathbf{0}$, as desired.

Now calculate:

$$\begin{aligned}\frac{1}{2}\dot{V} &= \frac{1}{\sigma}x\dot{x} + y\dot{y} + z\dot{z} \\ &= (yx - x^2) + (ryx - y^2 - xyz) + (zxy - bz^2) \\ &= (r+1)xy - x^2 - y^2 - bz^2.\end{aligned}$$

Completing the square in the first two terms gives

$$\frac{1}{2}\dot{V} = -\left[x - \frac{r+1}{2}y\right]^2 - \left[1 - \left(\frac{r+1}{2}\right)^2\right]y^2 - bz^2.$$

We claim that the right-hand side is strictly negative if $r < 1$ and $(x, y, z) \neq (0, 0, 0)$. It is certainly not positive, since it is a negative sum of squares. But could $\dot{V} = 0$?

That would require each of the terms on the right to vanish separately. Hence $y = 0$, $z = 0$, from the second two terms on the right-hand side. (Because of the assumption $r < 1$, the coefficient of y^2 is nonzero.) Thus the first term reduces to $-x^2$, which vanishes only if $x = 0$.

The upshot is that $\dot{V} = 0$ implies $(x, y, z) = (0, 0, 0)$. Otherwise $\dot{V} < 0$. Hence the claim is established, and therefore the origin is globally stable for $r < 1$.

Stability of C^+ and C^-

Now suppose $r > 1$, so that C^+ and C^- exist. The calculation of their stability is left as Exercise 9.2.1. It turns out that they are linearly stable for

$$1 < r < r_H = \frac{\sigma(\sigma + b + 3)}{\sigma - b - 1}$$

(assuming also that $\sigma - b - 1 > 0$). We use a subscript H because C^+ and C^- lose stability in a Hopf bifurcation at $r = r_H$.

What happens immediately after the bifurcation, for r slightly greater than r_H ? You might suppose that C^+ and C^- would each be surrounded by a small stable limit cycle. That would occur if the Hopf bifurcation were supercritical. But actually it's *subcritical*—the limit cycles are *unstable* and exist only for $r < r_H$. This requires a difficult calculation; see Marsden and McCracken (1976) or Drazin (1992, Q8.2 on p. 277).

Here's the intuitive picture. For $r < r_H$ the phase portrait near C^+ is shown schematically in Figure 9.2.4.

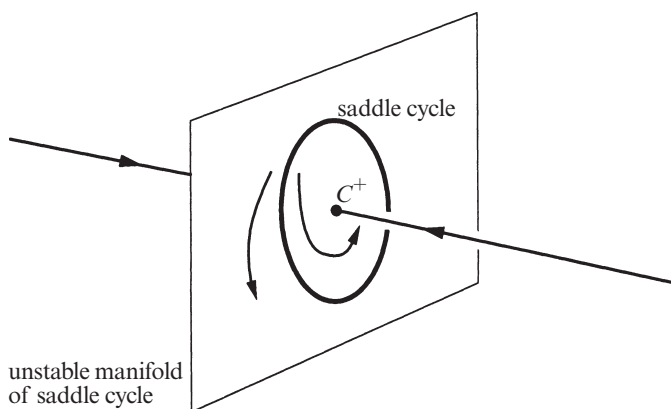


Figure 9.2.4

The fixed point is stable. It is encircled by a *saddle cycle*, a new type of unstable limit cycle that is possible only in phase spaces of three or more dimensions. The cycle has a two-dimensional unstable manifold (the sheet in Figure 9.2.4), and a two-dimensional stable manifold (not shown). As $r \rightarrow r_H$ from below, the cycle

shrinks down around the fixed point. At the Hopf bifurcation, the fixed point absorbs the saddle cycle and changes into a saddle point. For $r > r_H$ there are no attractors in the neighborhood.

So for $r > r_H$ trajectories must fly away to a distant attractor. But what can it be? A partial bifurcation diagram for the system, based on the results so far, shows no hint of any stable objects for $r > r_H$ (Figure 9.2.5).

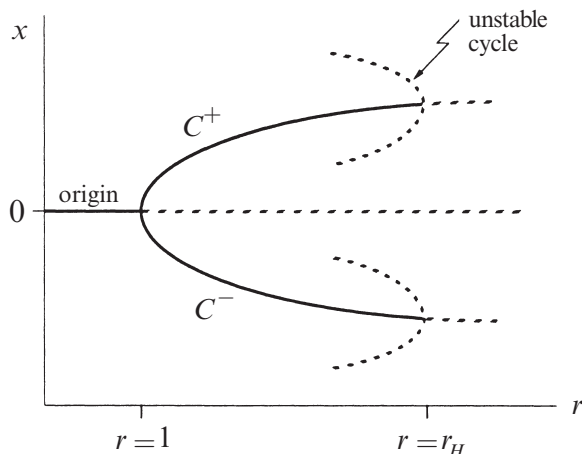


Figure 9.2.5

Could it be that all trajectories are repelled out to infinity? No; we can prove that all trajectories eventually enter and remain in a certain large ellipsoid (Exercise 9.2.2). Could there be some stable limit cycles that we're unaware of? Possibly, but Lorenz gave a persuasive argument that for r slightly greater than r_H , any limit cycles would have to be *unstable* (see Section 9.4).

So the trajectories must have a bizarre kind of long-term behavior. Like balls in a pinball machine, they are repelled from one unstable object after another. At the same time, they are confined to a bounded set of zero volume, yet they manage to move on this set forever without intersecting themselves or others.

In the next section we'll see how the trajectories get out of this conundrum.

9.3 Chaos on a Strange Attractor

Lorenz used numerical integration to see what the trajectories would do in the long run. He studied the particular case $\sigma = 10$, $b = \frac{8}{3}$, $r = 28$. This value of r is just past the Hopf bifurcation value $r_H = \sigma(\sigma + b + 3)/(\sigma - b - 1) \approx 24.74$, so he knew that something strange had to occur. Of course, strange things could occur for another reason—the electromechanical computers of those days were unreliable and difficult to use, so Lorenz had to interpret his numerical results with caution.

Then $x_{n+2} = f(f(x_n))$ or more simply, $x_{n+2} = f^2(x_n)$. Similarly, $x_{n+3} = f^3(x_n)$.

The third-iterate map $f^3(x)$ is the key to understanding the birth of the period-3 cycle. Any point p in a period-3 cycle repeats every three iterates, by definition, so such points satisfy $p = f^3(p)$ and are therefore fixed points of the third-iterate map. Unfortunately, since $f^3(x)$ is an eighth-degree polynomial, we cannot solve for the fixed points explicitly. But a graph provides sufficient insight. Figure 10.4.1 plots $f^3(x)$ for $r = 3.835$.

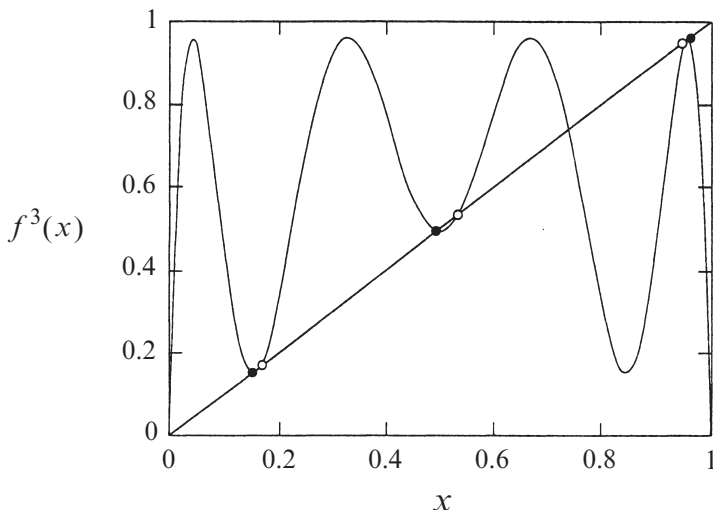


Figure 10.4.1

Intersections between the graph and the diagonal line correspond to solutions of $f^3(x) = x$. There are eight solutions, six of interest to us and marked with dots, and two imposters that are not genuine period-3; they are actually fixed points, or period-1 points for which $f(x^*) = x^*$. The black dots in Figure 10.4.1 correspond to a stable period-3 cycle; note that the slope of $f^3(x)$ is shallow at these points, consistent with the stability of the cycle. In contrast, the slope exceeds 1 at the cycle marked by the open dots; this 3-cycle is therefore unstable.

Now suppose we decrease r toward the chaotic regime. Then the graph in Figure 10.4.1 changes shape—the hills move down and the valleys rise up. The curve therefore pulls away from the diagonal. Figure 10.4.2 shows that when $r = 3.8$, the six marked intersections have vanished. Hence, for some intermediate value between $r = 3.8$ and $r = 3.835$, the graph of $f^3(x)$ must have become *tangent* to the diagonal. At this critical value of r , the stable and unstable period-3 cycles coalesce and annihilate in a **tangent bifurcation**. This transition defines the beginning of the periodic window.

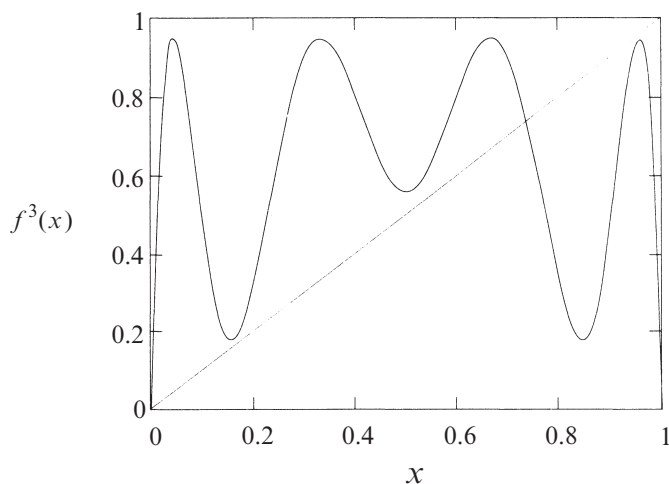


Figure 10.4.2

One can show analytically that the value of r at the tangent bifurcation is $1 + \sqrt{8} = 3.8284\dots$ (Myrberg 1958). This beautiful result is often mentioned in textbooks and articles—but always without proof. Given the resemblance of this result to the $1 + \sqrt{6}$ encountered in Example 10.3.3, I'd always assumed it should be comparably easy to derive, and once assigned it as a routine homework problem. Oops! It turns out to be a bear. See Exercise 10.4.10 for hints, and Saha and Strogatz (1994) for Partha Saha's solution, the most elementary one my class could find. Maybe you can do better; if so, let me know!

Intermittency

For r just below the period-3 window, the system exhibits an interesting kind of chaos. Figure 10.4.3 shows a typical orbit for $r = 3.8282$.

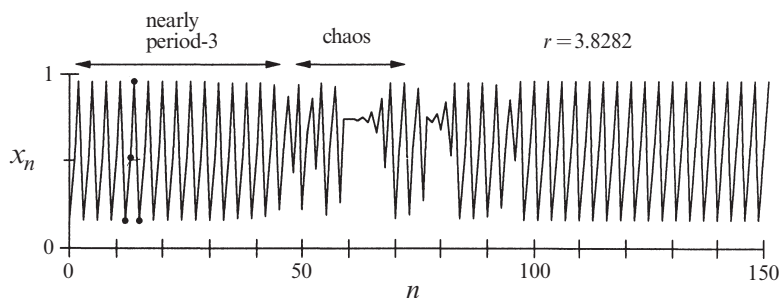


Figure 10.4.3

Part of the orbit looks like a stable 3-cycle, as indicated by the black dots. But this is spooky since the 3-cycle no longer exists! We're seeing the *ghost* of the 3-cycle.

We should not be surprised to see ghosts—they *always* occur near saddle-node bifurcations (Sections 4.3 and 8.1) and indeed, a tangent bifurcation is just a saddle-node bifurcation by another name. But the new wrinkle is that the orbit returns to the ghostly 3-cycle repeatedly, with intermittent bouts of chaos between visits. Accordingly, this phenomenon is known as *intermittency* (Pomeau and Manneville 1980).

Figure 10.4.4 shows the geometry underlying intermittency.

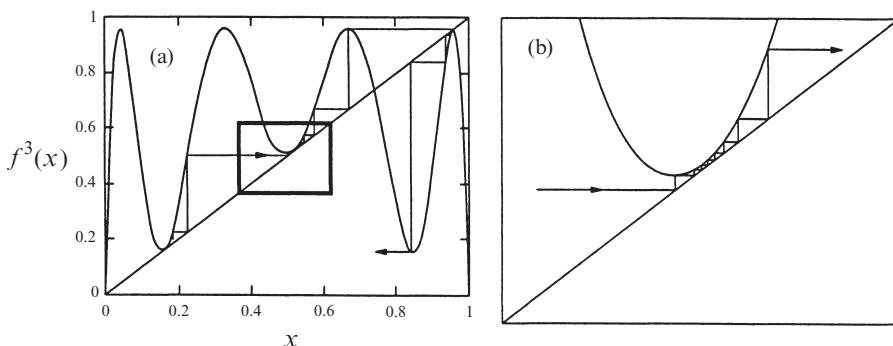


Figure 10.4.4

In Figure 10.4.4a, notice the three narrow channels between the diagonal and the graph of $f^3(x)$. These channels were formed in the aftermath of the tangent bifurcation, as the hills and valleys of $f^3(x)$ pulled away from the diagonal. Now focus on the channel in the small box of Figure 10.4.4a, enlarged in Figure 10.4.4b. The orbit takes many iterations to squeeze through the channel. Hence $f^3(x_n) \approx x_n$ during the passage, and so the orbit looks like a 3-cycle; this explains why we see a ghost.

Eventually, the orbit escapes from the channel. Then it bounces around chaotically until fate sends it back into a channel at some unpredictable later time and place.

Intermittency is not just a curiosity of the logistic map. It arises commonly in systems where the transition from periodic to chaotic behavior takes place by a saddle-node bifurcation of cycles. For instance, Exercise 10.4.8 shows that intermittency can occur in the Lorenz equations. (In fact, it was discovered there; see Pomeau and Manneville 1980).

In experimental systems, intermittency appears as nearly periodic motion interrupted by occasional irregular bursts. The time between bursts is statistically distributed, much like a random variable, even though the system is completely deterministic. As the control parameter is moved farther away from the periodic window, the bursts become more frequent until the system is fully chaotic. This progression is known as the *intermittency route to chaos*.

Figure 10.4.5 shows an experimental example of the intermittency route to chaos in a laser.

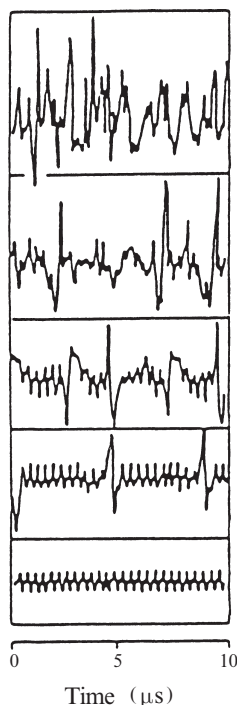


Figure 10.4.5

The intensity of the emitted laser light is plotted as a function of time. In the lowest panel of Figure 10.4.5, the laser is pulsing periodically. A bifurcation to intermittency occurs as the system's control parameter (the tilt of the mirror in the laser cavity) is varied. Moving from bottom to top of Figure 10.4.5, we see that the chaotic bursts occur increasingly often.

For a nice review of intermittency in fluids and chemical reactions, see Bergé et al. (1984). Those authors also review two other types of intermittency (the kind considered here is *Type I intermittency*) and give a much more detailed treatment of intermittency in general.

Period-Doubling in the Window

We commented at the end of Section 10.2 that a copy of the orbit diagram appears in miniature in the period-3 window. The explanation has to do with hills and valleys again. Just after the stable 3-cycle is created in the tangent bifurcation, the slope at the black dots in Figure 10.4.1 is close to $+1$. As we increase r , the hills rise and the valleys sink. The slope of $f^3(x)$ at the black dots decreases steadily